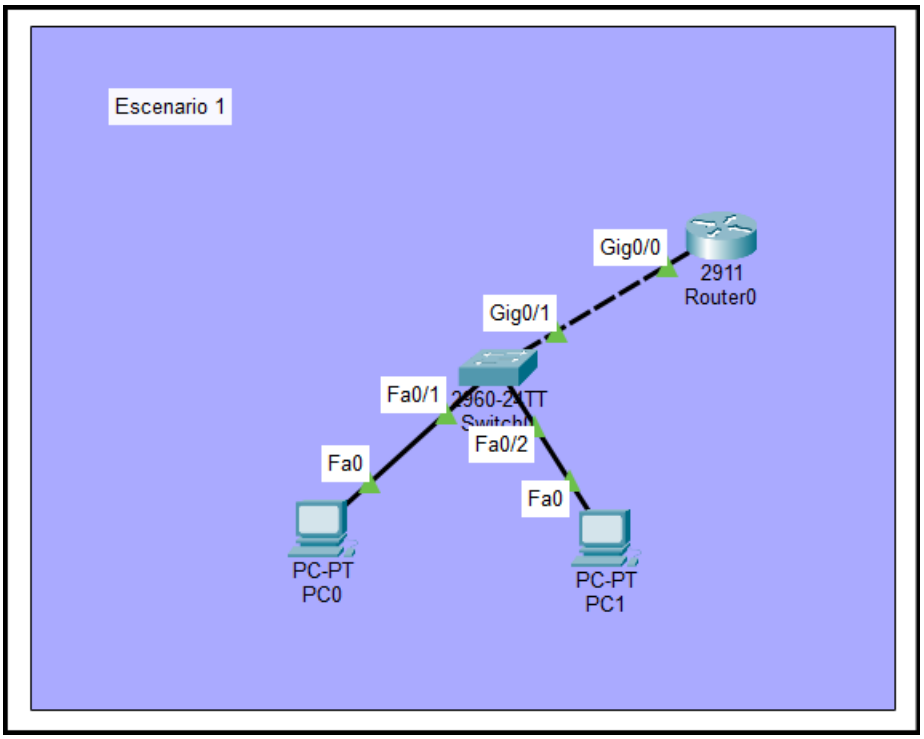
201701160
María Zucely Hernández García

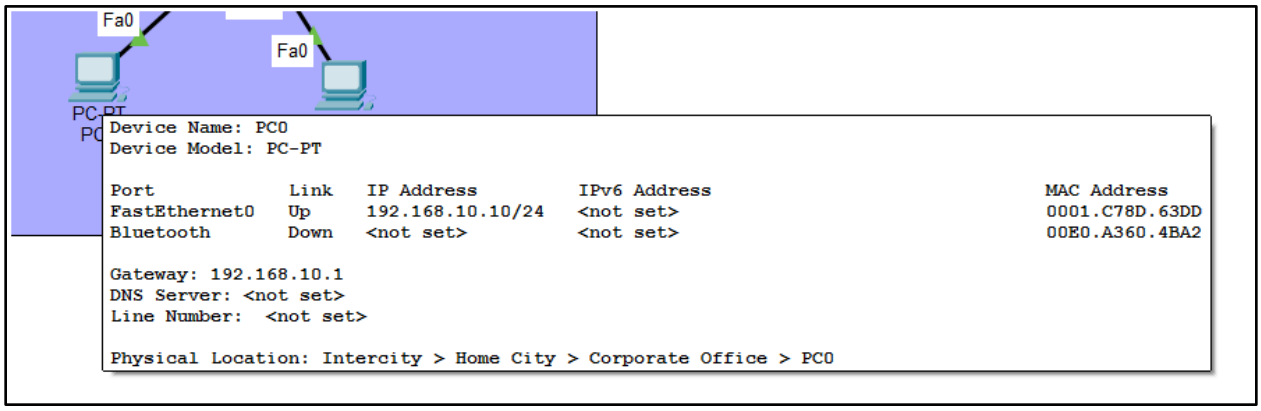
ACTIVIDAD PRÁCTICA 5

Escenario 1

- Topología

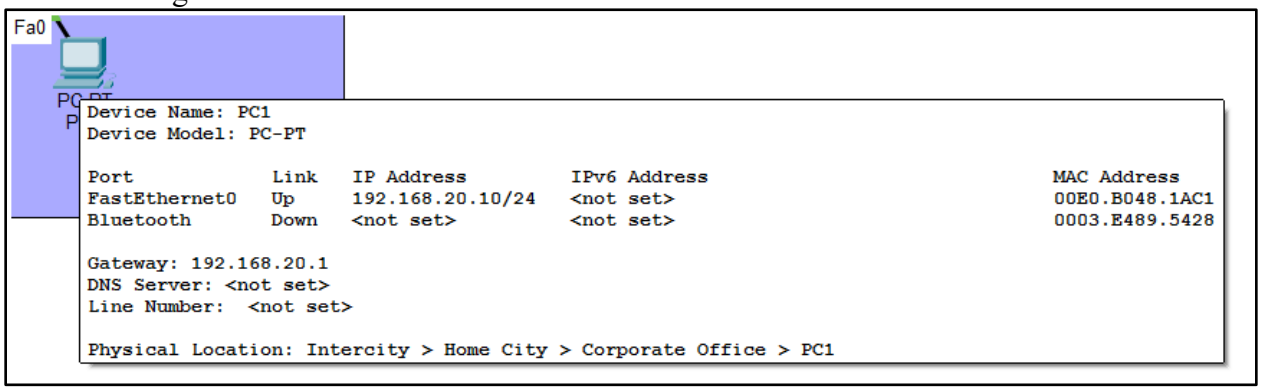


- Configuración IP de la PC0



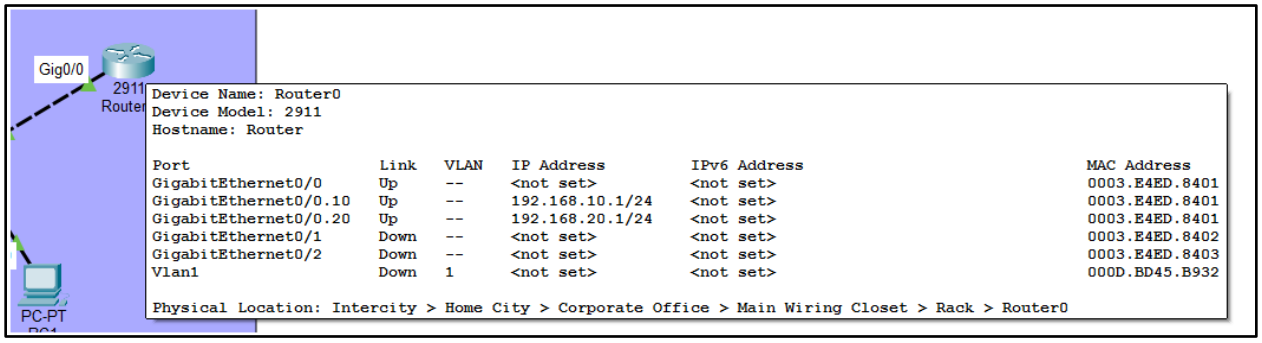
IP Address: 192.168.10.30
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

- Configuración IP de la PC1

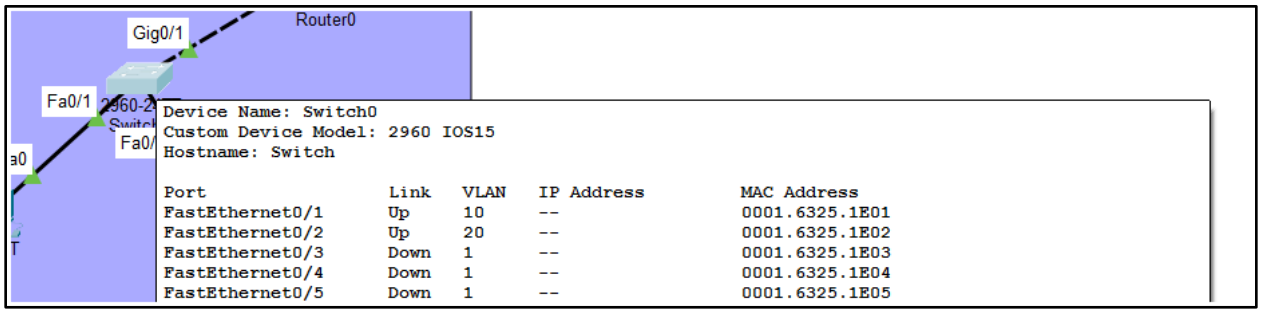


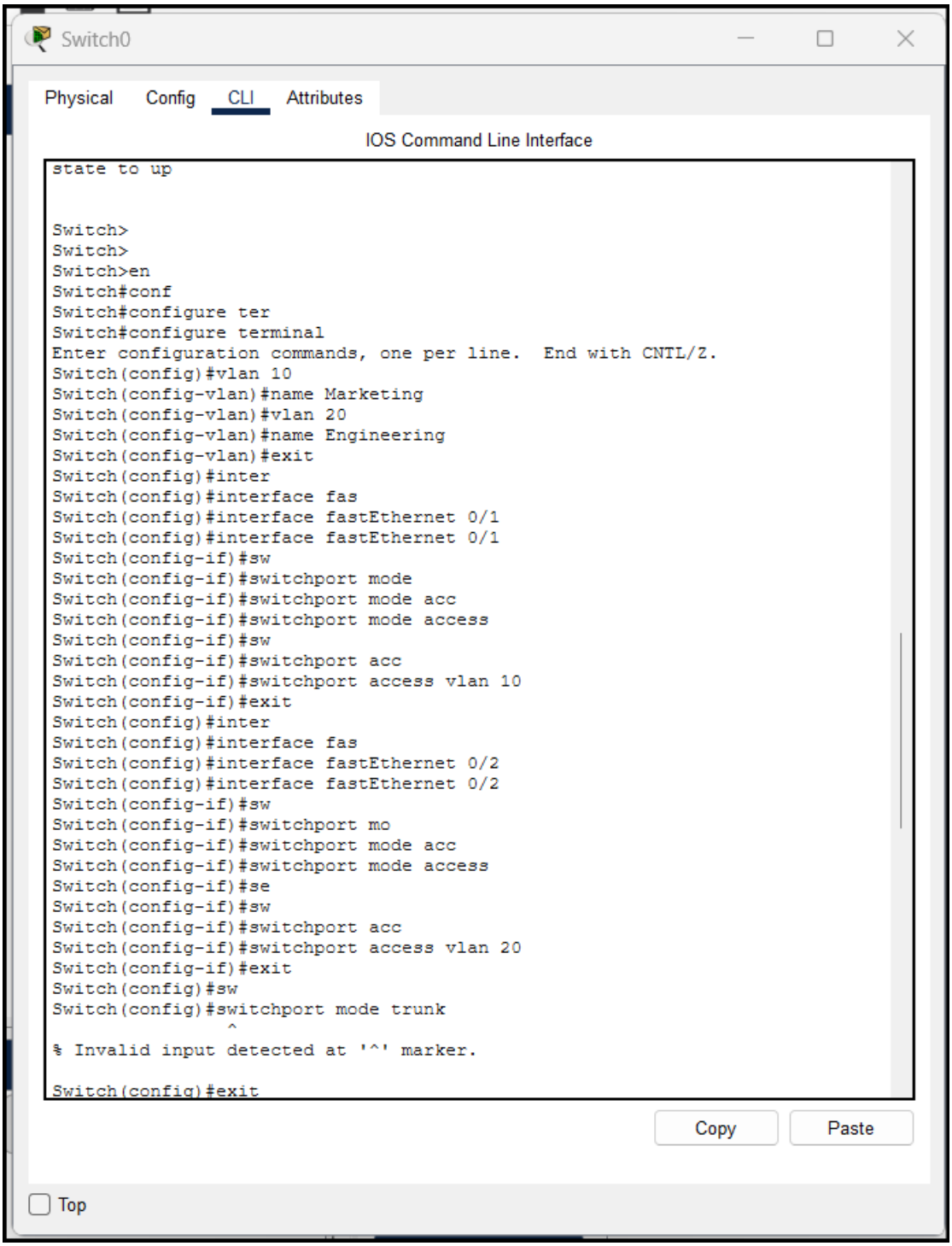
IP Address: 192.168.20.40
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

- Configuración del Router

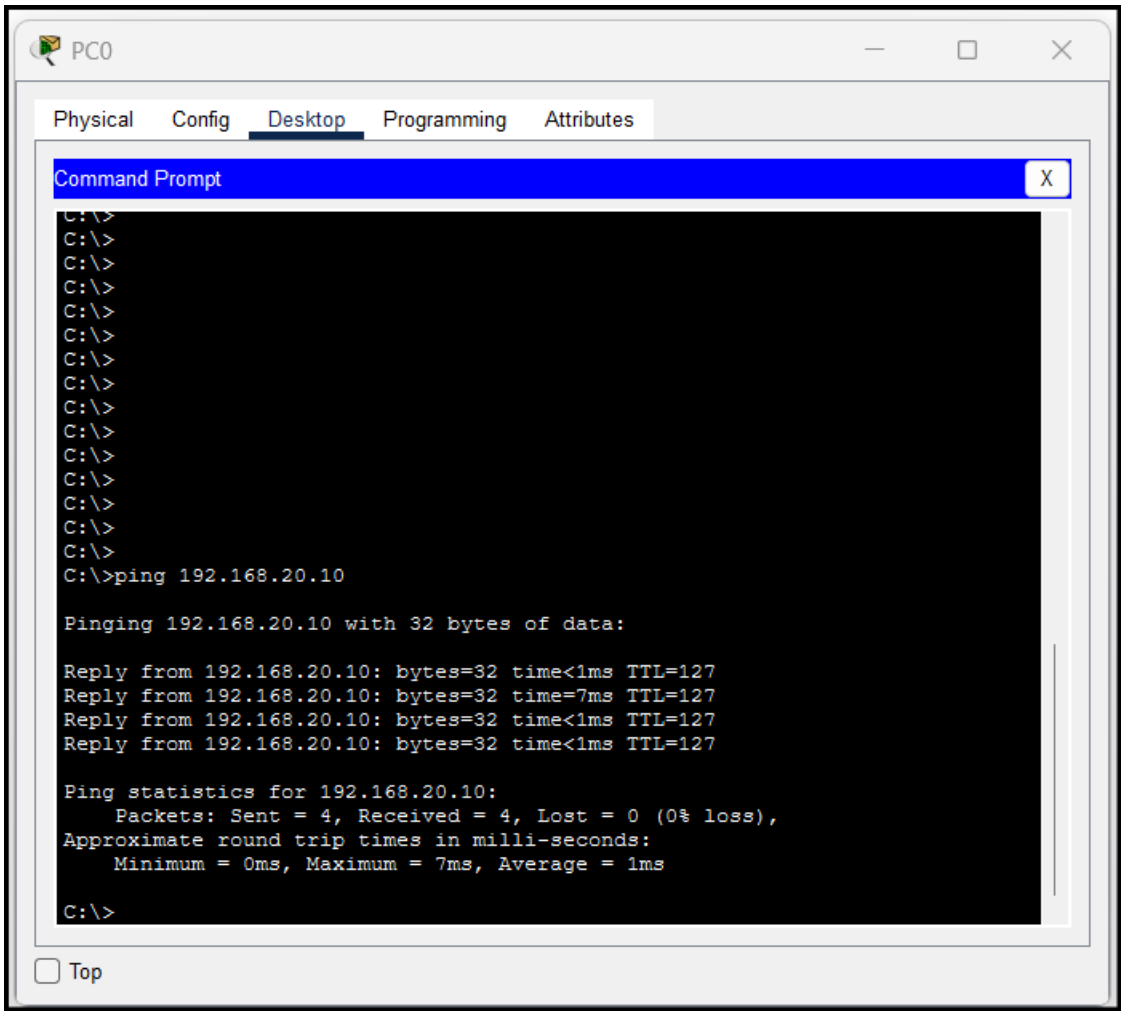


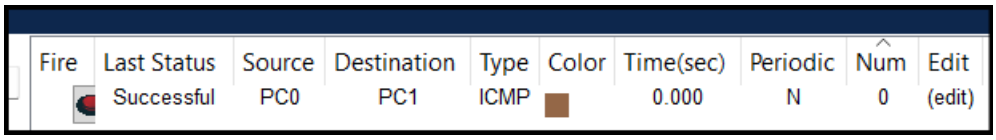
- Switch 0



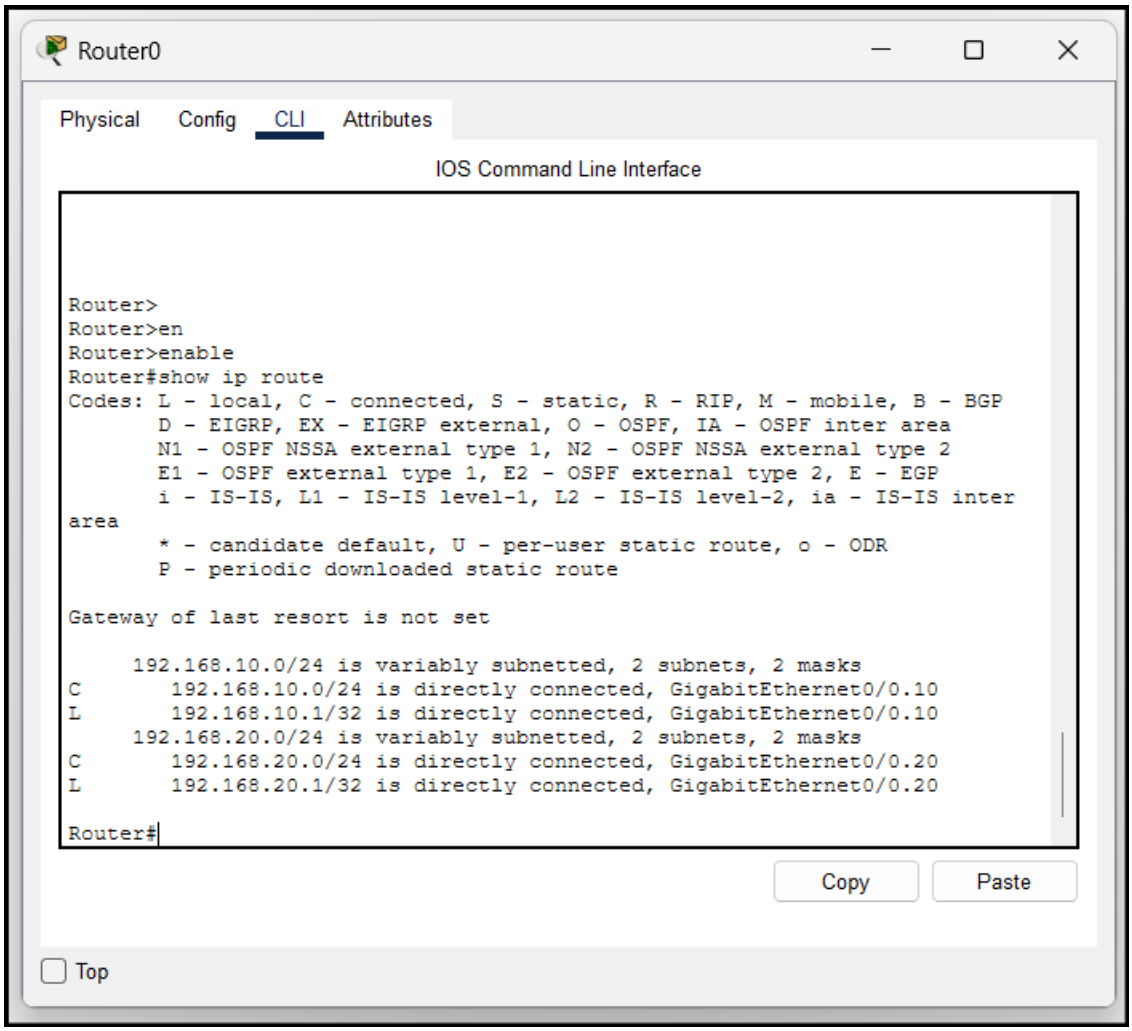

Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
state to up
Switch>
Switch>
Switch>en
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Marketing
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name Engineering
Switch(config-vlan)#exit
Switch(config)#inter
Switch(config)#interface fas
Switch(config)#interface fastEthernet 0/1
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#sw
Switch(config-if)#switchport mode
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#sw
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#inter
Switch(config)#interface fas
Switch(config)#interface fastEthernet 0/2
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#sw
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode acc
Switch(config-if)#switchport mode access
Switch(config-if)#se
Switch(config-if)#sw
Switch(config-if)#switchport acc
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#sw
Switch(config)#switchport mode trunk
^
% Invalid input detected at '^' marker.
Switch(config)#exit
Copy
Paste
Top

- Ping entre PC's



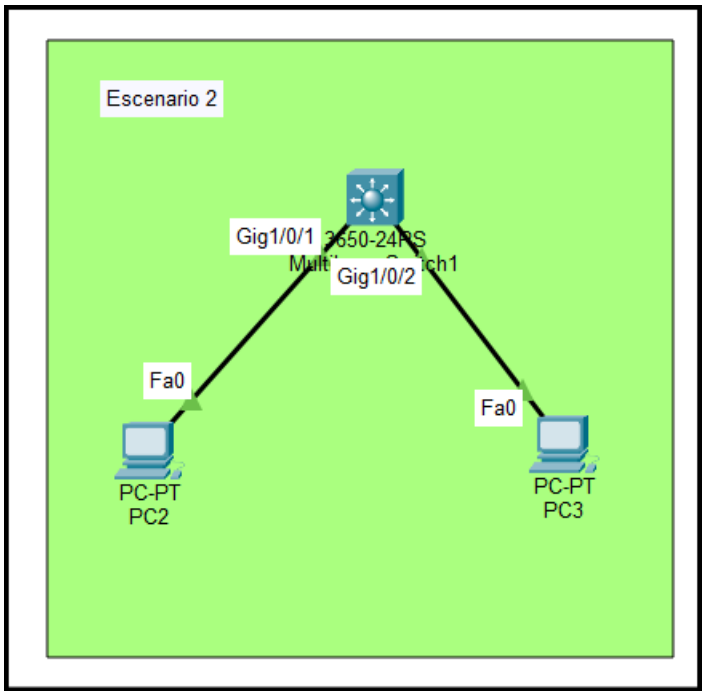


- Show ip route del Router

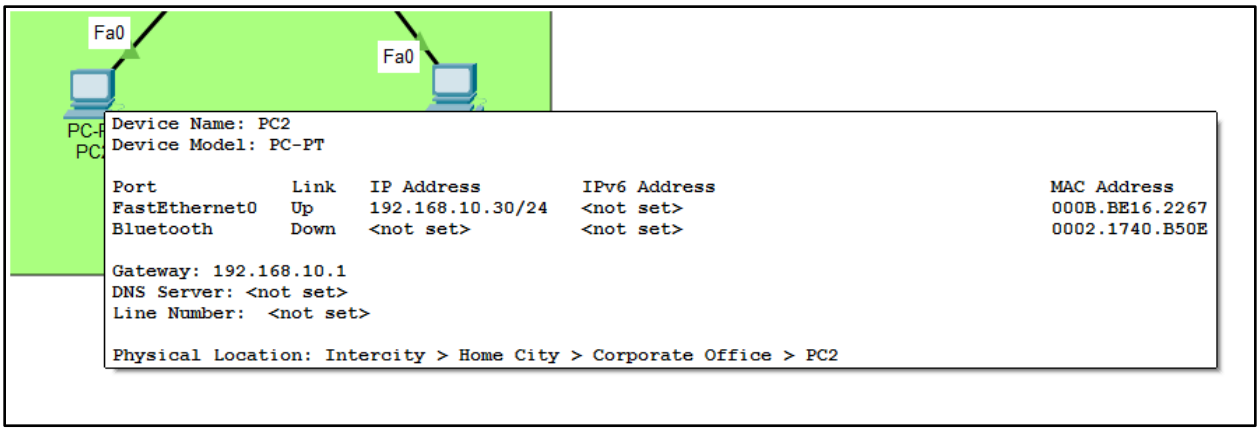


Escenario 2

- Topologia

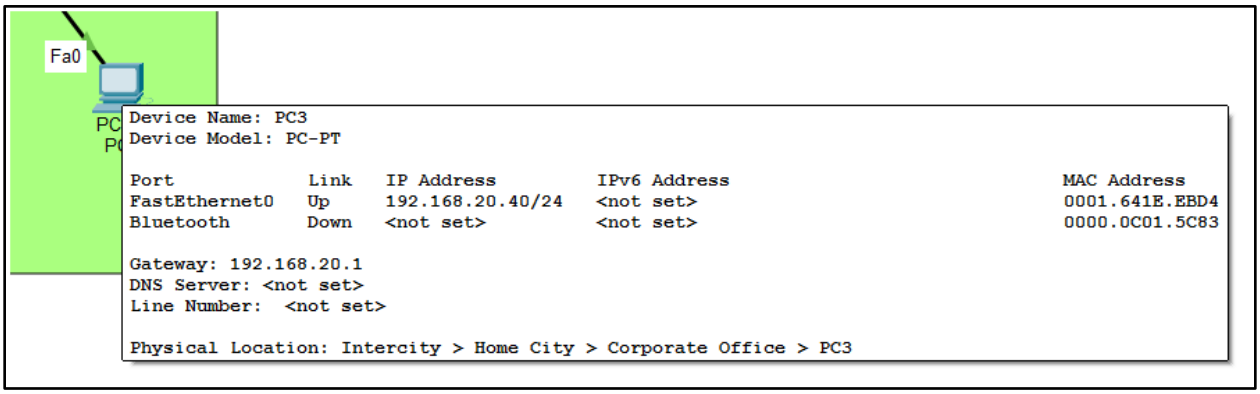


- Configuracion PC2



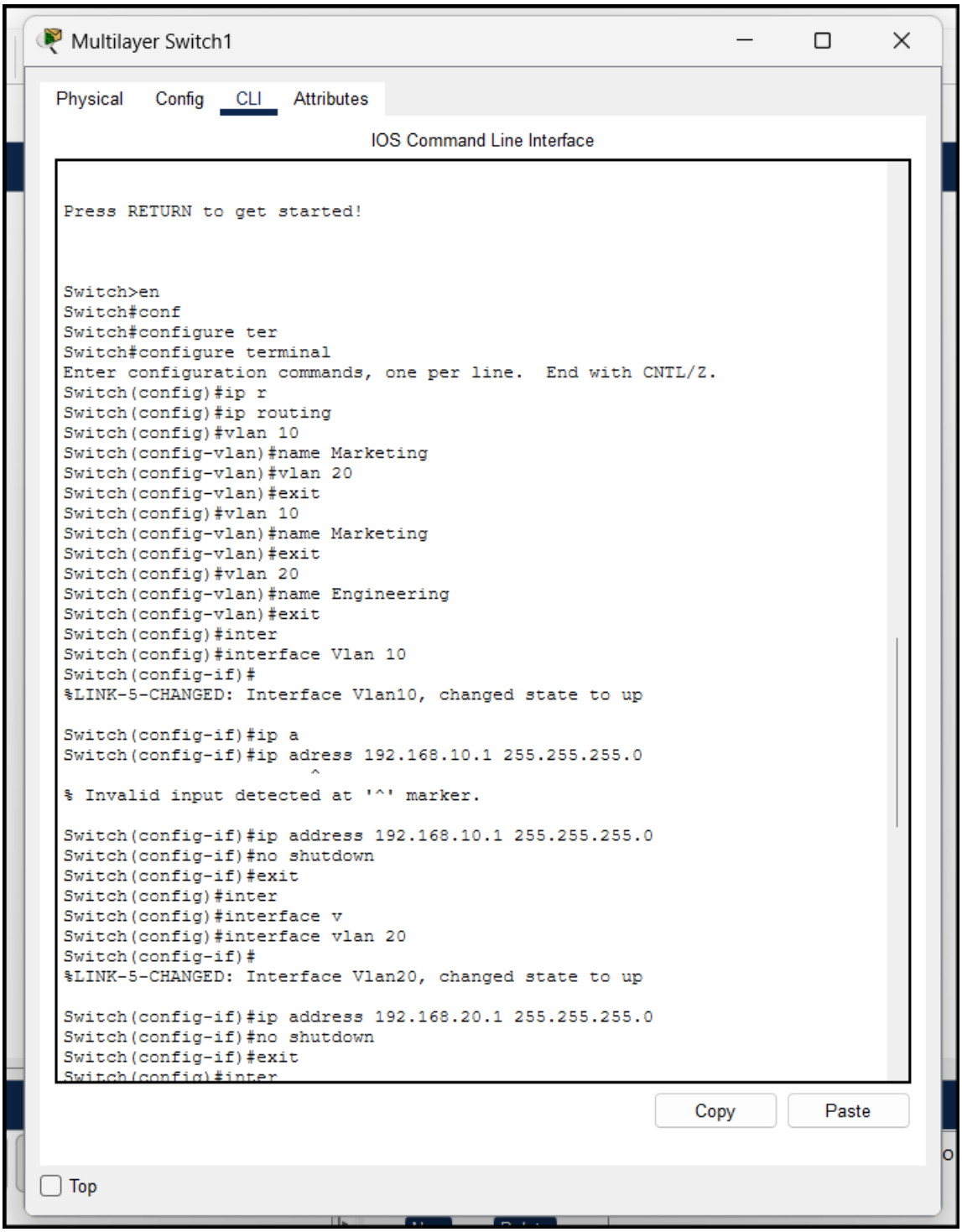
IP Address: `192.168.10.30`
Subnet Mask: `255.255.255.0`
Default Gateway: `192.168.10.1`

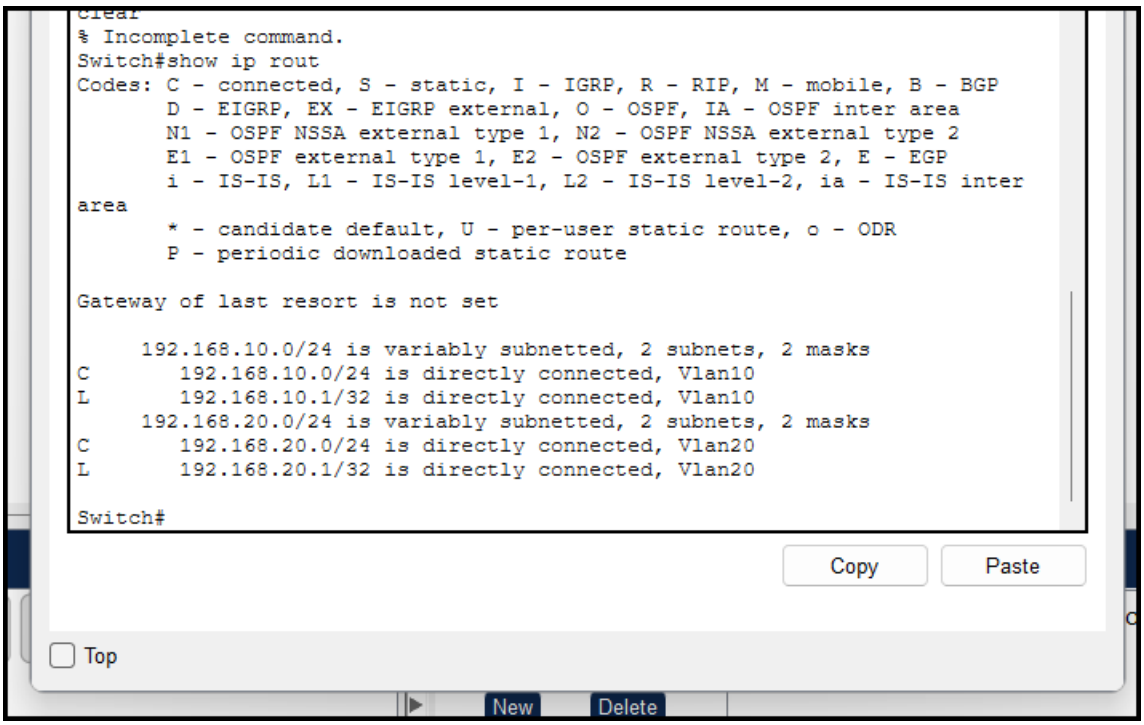
- Configuracion PC3

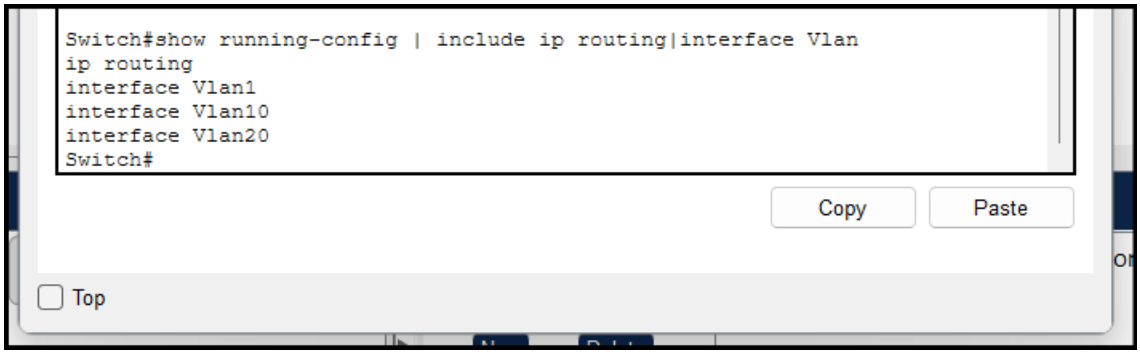


IP Address: 192.168.20.40
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

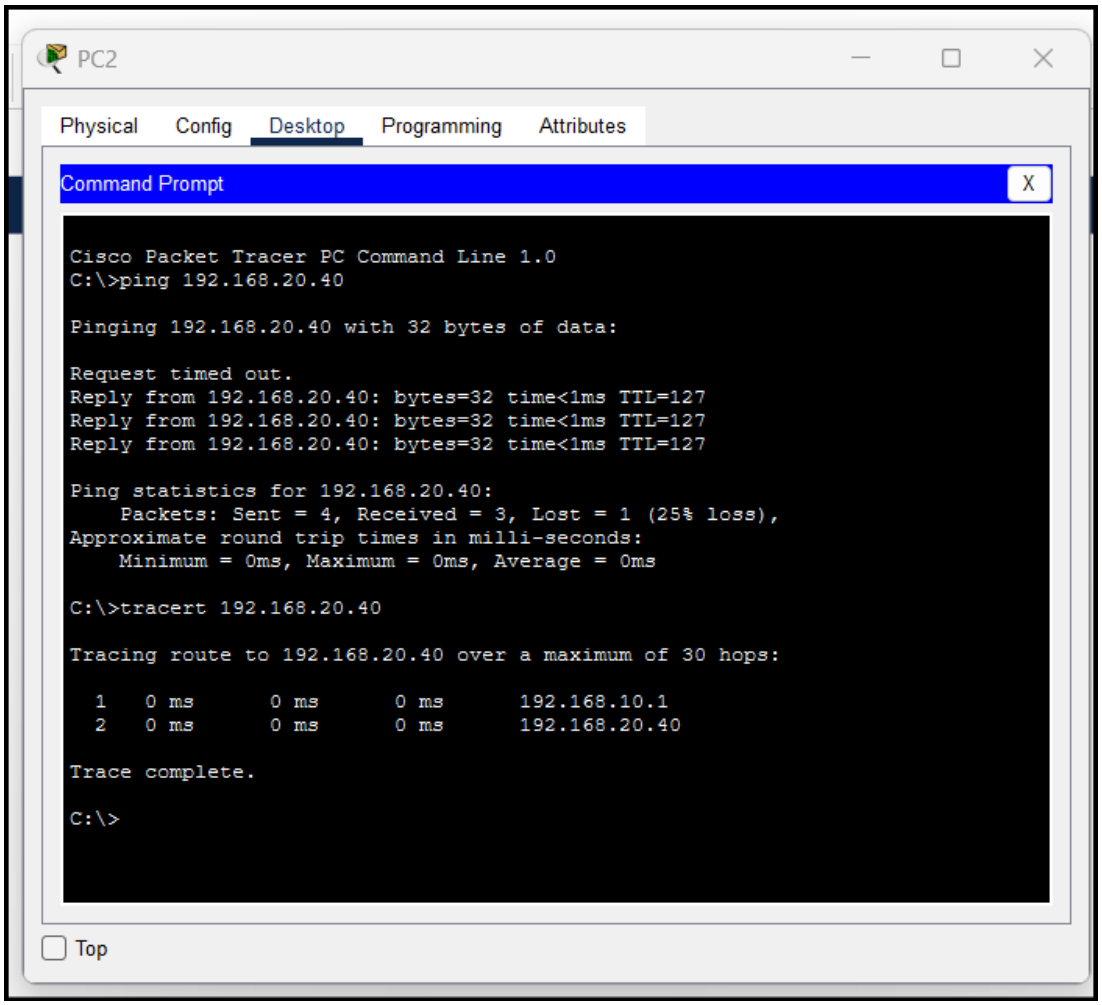
- Switch L3

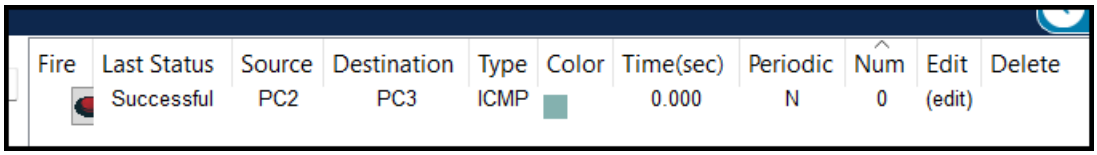

Multilayer Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
Switch>en
Switch#conf
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip r
Switch(config)#ip routing
Switch(config)#vlan 10
Switch(config-vlan)#name Marketing
Switch(config-vlan)#vlan 20
Switch(config-vlan)#exit
Switch(config)#vlan 10
Switch(config-vlan)#name Marketing
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Engineering
Switch(config-vlan)#exit
Switch(config)#inter
Switch(config)#interface Vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up
Switch(config-if)#ip a
Switch(config-if)#ip adress 192.168.10.1 255.255.255.0
^
% Invalid input detected at '^' marker.
Switch(config-if)#ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#inter
Switch(config)#interface v
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up
Switch(config-if)#ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#inter
Copy
Paste
Top


clear
% Incomplete command.
Switch#show ip rout
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, Vlan10
L 192.168.10.1/32 is directly connected, Vlan10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.20.0/24 is directly connected, Vlan20
L 192.168.20.1/32 is directly connected, Vlan20
Switch#
Copy
Paste
Top
New
Delete


Switch#show running-config | include ip routing|interface Vlan
ip routing
interface Vlan1
interface Vlan10
interface Vlan20
Switch#
Copy
Paste
Top

- Ping entre Vlans





- ¿Cuál de los dos métodos considera más eficiente para una red empresarial de alto tráfico y por qué?
  El método más eficiente sería usando el switch multicapa ya que:
  Es mucho más rápido: El Switch de Capa 3 procesa los datos por chip (hardware), lo que hace que la comunicación entre VLANs sea casi instantánea. El router, en cambio, lo hace por software y es más lento.
  Evita embotellamientos: En el método con Router (Escenario 1), todo el tráfico tiene que pasar a la fuerza por un solo cable hacia el router, lo que satura la red. En el Switch Capa 3, todo el intercambio se hace internamente dentro del mismo equipo.